

Roland Piquepaille's Technology Trends

How new technologies are modifying our way of life



mercredi 7 avril 2004

Social Networking Tools Applied to Scientific Topics

In "**Mapping the landscape of science**," the National Science Foundation discusses the contents of a collection of articles published by the *Proceedings of the National Academy of Sciences* (PNAS), "**Mapping Knowledge Domains**." Basically, all these scientists are using software social networking tools to build graphical representations of scientific knowledge or science communities. [Please note that the full version of all articles is available.]

The results featured in PNAS were originally presented at the May 2003 Arthur M. Sackler Colloquium on Mapping Knowledge Domains, sponsored by the National Academy of Sciences. Organized by Richard Shiffrin and Börner of Indiana University, the colloquium addressed the task of extracting meaningful and relevant information from largely unorganized data collections.

"Today, almost all of us access knowledge in ways vastly different from those used for hundreds of years," Shiffrin said. "The traditional method involved books, reference works and physical materials on library shelves, most of which had been verified for accuracy by one or another authority. Now, we sit at computers and cast our net into a sea of information, much of which is inaccurate or misleading."

This explains why most of the authors turned to software to map scientific knowledge. Here is an example.

In **his paper**, Mark Newman showed that clusters in social networks can also be used to map scientific communities. A scientist may or may not be six degrees from Kevin Bacon, but Newman showed that scientists were about six coauthors away from any other scientist.

Below are some illustrations and their legends extracted from some of this collection of scientific articles.

This one comes from a paper by Ketan K. Mane and Katy Börner, "**Mapping topics and topic bursts in PNAS**." It shows the "co-word space of the top 50 highly frequent and bursty words used in the top 10% most highly cited PNAS publications in 1982-2001."

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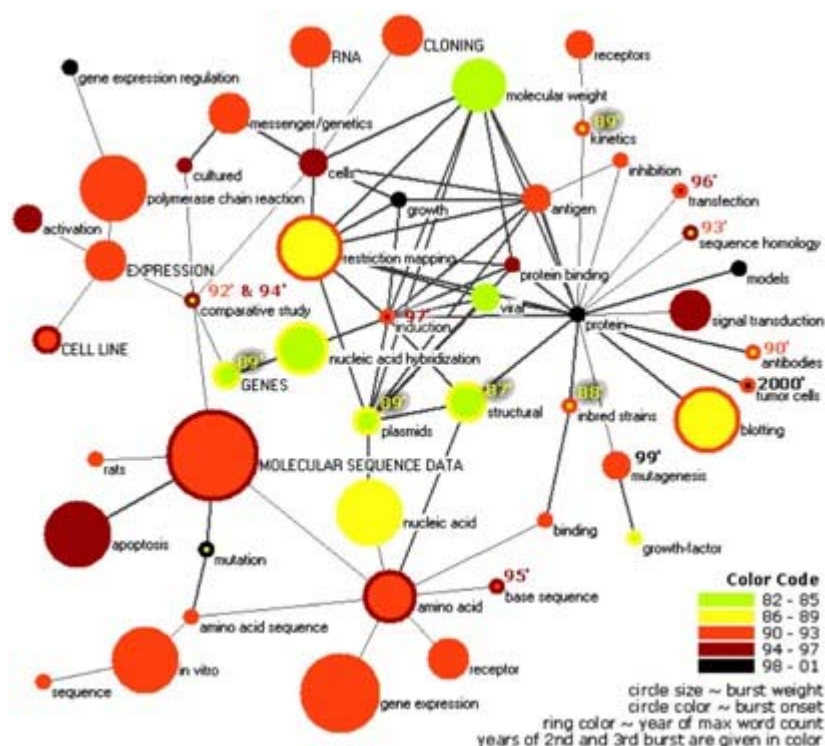
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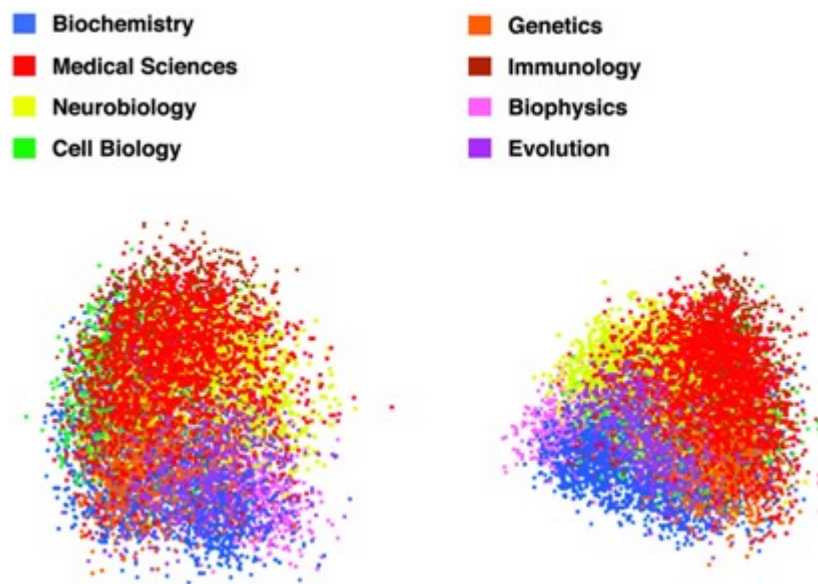
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And this one comes from a paper by Thomas K. Landauer, Darrell Laham and Marcia Derr, "[From paragraph to graph: Latent semantic analysis for information visualization](#)." It shows two maps of PNAS articles colored by biology subfield categories. The two-dimensional view on the three-dimensional space was selected algorithmically (left) and by aided human selection (right).



If you don't have enough information with this collection about twenty scientific articles, you might want to read some previous language-refreshing stories about social-network mapping tools, [here](#) or [there](#).

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